Week 9

Privacy-Preserving Federated Learning

Threat models, leakage, and defenses

Graduate course · English instruction



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The Privacy Misconception

Keeping raw data on the client reduces exposure, but model updates can still reveal information about the local training data.

Raw data local

A strong privacy-by-design starting point.

Updates shared

Gradients or model deltas may encode sensitive signals.

Threat model first

The right defense depends on who is trusted and what is exposed.

Local data storage alone does not establish a formal privacy guarantee.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Learning Objectives

Threat models

Distinguish outsider, curious client, and honest-but-curious / malicious server.

Inference attacks

Explain gradient reconstruction, membership inference, and property inference at a high level.

Differential privacy

Understand the meaning of ε, δ, and privacy–utility trade-offs.

Cryptographic defenses

Compare secure aggregation, homomorphic encryption, and multi-party computation.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Who Can Be the Adversary?

External attacker

May query a released model or target communication infrastructure.

Curious client

Participates legitimately but tries to infer information about other clients.

Curious server

Follows the protocol but inspects individual updates or intermediate statistics.

Malicious participant

Actively deviates from the protocol to extract information or change the model.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Main Inference Threat Families

Training-data reconstruction

Recover input examples or sensitive features from gradients / u pdates.

Membership inference

Determine whether a particular record contributed to training.

Model inversion

Construct representative inputs associated with model outputs or classes.

Property inference

Infer attributes of a client dataset unrelated to the main learnin g task.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Gradient Leakage: High-Level Attack Intuition

Why gradient leakage is possible

Private batch

Client computes a loss and gradient s from local data.

→

Shared gradient

The server observes an update relat ed to the client data.

→

Reconstruction

An attacker searches for dummy inp uts whose gradients match the obse rved update.

“No raw data transfer” does not imply “no information leakage.”

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Privacy Must Be Defined, Not Assumed

A privacy statement should specify the adversary, the information visible to that adversary, and the informati on the defense is designed to protect.

Input privacy

Protect individual client updates.

Output privacy

Protect information exposed by the learned model.

End-to-end design

Different mechanisms protect diffe rent attack surfaces.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Differential Privacy

$$\Pr[\boldsymbol{M}(\boldsymbol{D}) \in \boldsymbol{S}] \leq \boldsymbol{e}^{\boldsymbol{\varepsilon}} \cdot \Pr[\boldsymbol{M}(\boldsymbol{D}') \in \boldsymbol{S}] + \boldsymbol{\delta}$$

For neighboring datasets D and D' that differ in one protected unit, the output distributions should remain close. Smaller ε generally means stronger privacy.

ε

Privacy loss parameter.

δ

Small additive slack in the privacy bo und.

Mechanism

Usually combines clipping with rando mized noise.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Client-Level DP in Federated Learning

1. Local update

Client computes an update.

→

2. Clip

Bound the influence of each client update.

→

3. Aggregate

Combine selected client updates.

→

4. Noise

Add calibrated random noise.

→

5. Account

Track cumulative privacy loss across rounds.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

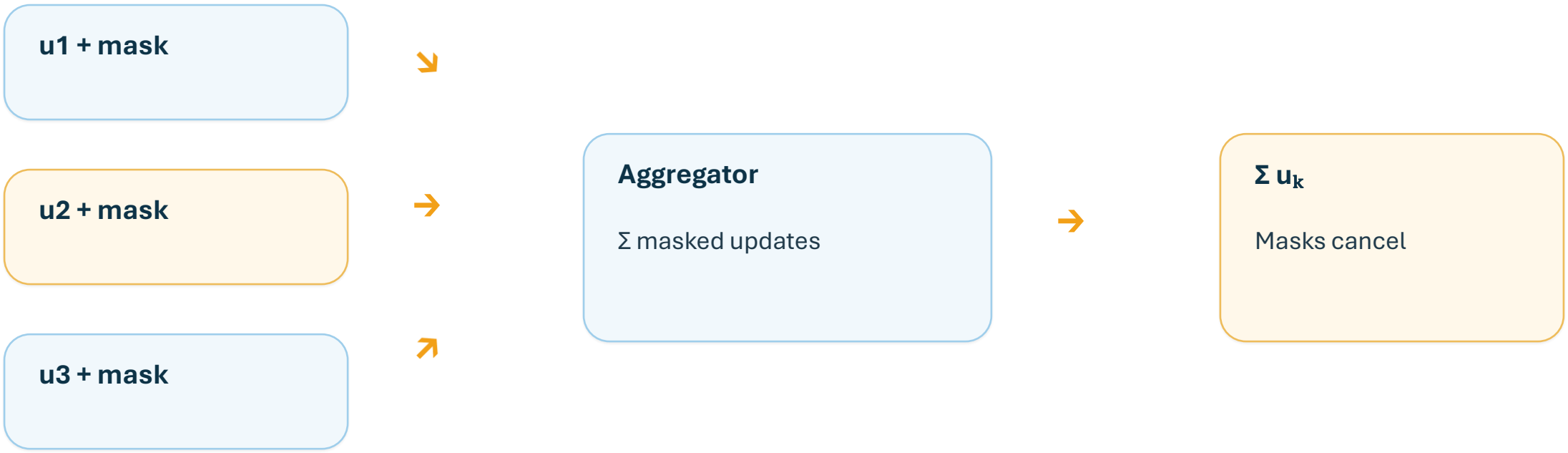
Local DP vs. Central / Global DP

Property	Local DP	Global / central DP
Who adds noise?	Each client before upload	Trusted aggregator after secure aggregation / collection
Trust in server	Low	Higher
Utility	Often lower due to more noise	Typically better for the same privacy target
Threat coverage	Protects client updates from server	Protects contribution in released aggregate/model

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Secure Aggregation: Protect Individual Updates

Secure aggregation: the server learns only an aggregate



Individual updates remain hidden under the protocol's trust and participation assumptions.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Homomorphic Encryption and MPC

Homomorphic encryption

Supports computation over encrypted updates. Additive HE is naturally aligned with weighted summation but can be computationally expensive.

Secure multi-party computation

Distributes trust across multiple participants or servers so no single entity sees all private inputs.

Design choice

Cryptography changes latency, message complexity, supported aggregation rules, and dropout handling.

Important limitation

Secure aggregation protects individual updates. It does not automatically protect information leaked by the final model.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Privacy Mechanisms Protect Different Things

Mechanism	Protects updates from server?	Protects model outputs?	Main cost
Secure aggregation	Yes	No	Extra protocol / messages
Homomorphic encryption	Yes	No by itself	Heavy computation / ciphertext expansion
Differential privacy	Depends on placement	Yes, for defined contribution	Accuracy / privacy budget
Hybrid SA + DP	Yes	Yes, with assumptions	Combined overhead

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Privacy–Utility–Efficiency Trade-off

Stronger protection

More clipping/noise or heavier cryptography can reduce leakag e risk, but may decrease model utility or increase training cost.

Weaker protection

Lower overhead and potentially higher accuracy, but more infor mation may be exposed to the server, clients, or model users.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

In-Class Threat-Model Exercise

Scenario A

Hospitals trust a university server but not each other. What mec hanism is needed?

Scenario B

Mobile users do not trust the service provider with individual up dates. What changes?

Scenario C

The final global model is public. Which threats remain after sec ure aggregation?

Task

For each scenario: adversary → exposed artifact → threat → defe nse → cost.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Key Takeaways

What to remember before leaving today

1. Federated learning reduces raw-data movement but does not guarantee privacy by itself.
2. Threat modeling should precede the choice of a privacy mechanism.
3. Differential privacy provides a formal output-privacy guarantee under explicit parameters and assumptions.
4. Secure aggregation and cryptographic techniques primarily protect individual updates from the aggregator.
5. Privacy defenses must be evaluated together with model utility, communication, computation, and trust assumpti ons.

Ludwig & Baracaldo (2022), Chs. 13–15 · Kairouz et al. (2021), §4

Reading & Preparation

Suggested reading before the next session

Main textbook

Ludwig & Baracaldo (2022), Chapters 13–15 on leakage, aggrega tion privacy, and inference attacks.

Supplementary reference

Kairouz et al. (2021), Section 4: Preserving the Privacy of User D ata.

Preparation: distinguish privacy attacks from security attacks that manipulate model behavior.

Q & A

Thank You.

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